

$$\begin{aligned}
 = & - \sum_n (D_\mu \phi)_n^* (D^\mu \phi)_n - \frac{1}{2} \sum_n \left(\bar{\psi}_n \gamma^\mu (D_\mu \psi)_n \right) + \sum_n \mathcal{F}_n^* \mathcal{F}_n \\
 & - \text{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \left(\psi_{nL}^\top \epsilon \psi_{mL} \right) + 2 \text{Re} \sum_n \frac{\partial f(\phi)}{\partial \phi_n} \mathcal{F}_n \\
 & - 2\sqrt{2} \text{Im} \sum_{Anm} (t_A)_{nm} \left(\bar{\psi}_{nL} \lambda_A \right) \phi_m + 2\sqrt{2} \text{Im} \sum_{Anm} (t_A)_{mn} \left(\bar{\psi}_{nR} \lambda_A \right) \phi_m^* \\
 & - \sum_{Anm} \phi_n^* (t_A)_{nm} \phi_m D_A - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A \\
 & - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A \left(\bar{\lambda}_A (\not{p} \lambda)_A \right) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} . \quad (27.4.1)
 \end{aligned}$$

Here $f(\phi)$ is the superpotential, a gauge-invariant complex function of ϕ_n (but not of ϕ_n^*) which the condition of renormalizability requires to be a cubic polynomial; the ξ_A are constants which gauge invariance requires to vanish except where t_A is a $U(1)$ generator; the gauge-covariant derivatives are

$$D_\mu \psi_L \equiv \partial_\mu \psi_L - i \sum_A t_A V_{A\mu} \psi_L , \quad (27.4.2)$$

$$D_\mu \phi \equiv \partial_\mu \phi - i \sum_A t_A V_{A\mu} \phi , \quad (27.4.3)$$

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C , \quad (27.4.4)$$

and $f_{A\mu\nu}$ is the gauge-covariant gauge field-strength tensor

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu} . \quad (27.4.5)$$

The auxiliary fields enter quadratically, with field-independent constants as the coefficients of the second-order term, so they can be eliminated by setting them equal to the values at which the Lagrangian density is stationary:

$$\mathcal{F}_n = - \left(\partial f(\phi) / \partial \phi_n \right)^* , \quad (27.4.6)$$

$$D_A = \xi_A + \sum_{nm} \phi_n^* (t_A)_{nm} \phi_m . \quad (27.4.7)$$